- ٤هـ

6. رقم الكمبيوتر *

9

7. الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس) *

EZDH

✓ **Correct** 2/2 Points

8. By applying the construct heap algorithm on the next array, 12| 5| 20 |15 | 19 | 10 | 7 | 17 | 16 ,the resulting array will be *

- 20|19|12|17|5|10|7|15|16
- 20|19|12|17|5|10|16|15|7
- 20|19|12|17|5|10|7|16|15
- 20|19|12|17|7|10|5|15|16

✓ **Correct** 1/1 Points

9. f (n) = O(g(n)) implies g(n) = Ω (f (n)). *

- True
- False

✓ **Correct** 2/2 Points

10. The Θ expression (tight bound) to describe the number of operations required for the following algorithms is ……:

```
F2 (N)
    I = 1
    While I < N Do
        For J = 1 to I
          K = 1
        I = 2 × I
EndF2
```

*

- ○ $\Theta(N^2)$
- ◉ $\Theta(N)$
- ○ $\Theta(lgN)$
- ○ $\Theta(N \lg N)$

✓ **Correct** 1/1 Points

11. *

$T(n) = T(n/2) + 3T(n/2) + n^2 \ , T(n) = O(\ldots\ldots)$

- ◉ $n^2 \lg n$
- ○ n
- ○ n^2
- ○ $n \lg\ n$

✓ **Correct** 2/2 Points

12. The Θ expression (tight bound) to describe the number of operations required for the following algorithms is ……:

```
for (i = 2 ; i < N ; i *= 2)
  A[i] = i ;
```

*

- ○ $\Theta(N \lg N)$
- ◉ $\Theta(lgN)$
- ○ $\Theta(N^2)$
- ○ $\Theta(N)$

✓ **Correct** 2/2 Points

13. **T(n) = 2T(n/3) + n lg n , T(n)=O(……..)** *

$n \lg n$

Correct answers: $n \lg n$

✓ **Correct** 1/1 Points

14. *

$T(n) = 2T(n/2) + 4^{\lg n}$, $T(n) = O(\ldots\ldots)$

- ○ $n \lg \lg n$
- ○ $n \lg n$
- ○ n
- ◉ n^2

✓ **Correct** 2/2 Points

15. T(n) = 2*T(n-1) + c , T(n) = O(.....) *

2^n

✓ **Correct** 2/2 Points

16. **T(n) = T(n/5)+ T(4n/5) + n2 , T(n)=O(.....)** *

n^2

✓ **Correct** 1/1 Points

17. Binary search. **O(.....)** *

$\log\ n$

✓ **Correct** 2/2 Points

18. **T(n) = 2T(√n) + Q(lg lg n) , T(n)=O(.....)** *

$\lg\ n$

Correct answers: $\lg n$

✓ **Correct** 1/1 Points

19. Merge sort. **O(.....)** *

$n \log n$

✓ **Correct** 2/2 Points

20. By applying the quick sort on the next array, 56 | 82 | 92 | 74 | 76 | 68 | 88 | 68 | 48 | 74 | 18 | 94 , after the forth round of the partitioning the array will be like:
*

- (●) $[18, 48, 56, 68, 68, 74, 74, 76, 88, 82, 92, 94]$
- ( ) $[18, 48, 56, 74, 76, 68, 88, 68, 82, 74, 94, 92]$
- ( ) $[18, 48, 56, 68, 68, 74, 74, 76, 82, 88, 92, 94]$
- ( ) $[18, 48, 56, 74, 76, 68, 88, 68, 82, 74, 92, 94]$

✓ **Correct** 1/1 Points

21. *

$T(n) = T(n/5) + 4T(n/5) + n^2 \ , T(n) = O(\ldots\ldots)$

- ( ) n
- (●) n^2
- ( ) $n^2 \lg n$
- ( ) $n \lg n$

✗ **Incorrect** 0/1 Points

22. Any comparison based search has the lower bound of O(n).
*

False

True

✓ **Correct** 2/2 Points

23. The Θ expression (tight bound) to describe the number of operations required for the following algorithms is …….:

```
z = 0 ;
for (w = 0 ; w < K ; w++)
{
   for (x = 0 ; x < N ; x++)
     z = z + x ;
  for (y = 0 ; y < M ; y++)
     z = z + y ;
}
```

*

$b.\ \Theta(K \times (N + M))$

$\Theta(N \lg N)$

$\Theta(N^3)$

$\Theta(N^2)$

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